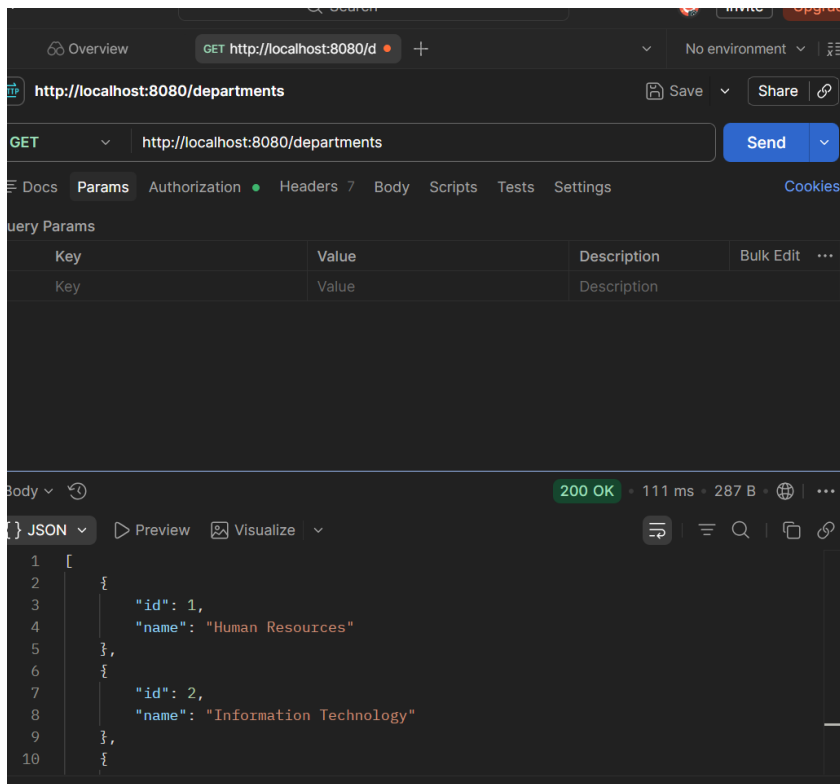


Exercise 1: Implementing Centralized Authentication with OAuth 2.1/OIDC

In this exercise, a centralized authentication system was implemented using **OAuth 2.1** and **OpenID Connect (OIDC)** in a Spring Boot application. The application was configured with **Spring Security** and the **OAuth2 Client** dependency to enable secure user authentication through **Google**. OAuth client properties, including the Client ID, Client Secret, scopes, and redirect URI, were configured in the `application.yml` file. A `SecurityConfig` class was created to secure all application endpoints and redirect unauthenticated users to the Google login page. After successful authentication, Google returned the user's identity information, and the application established an authenticated session. A `UserController` was implemented to retrieve and display the authenticated user's details, such as name, email, and profile information. This implementation demonstrates the concept of **Single Sign-On (SSO)**, where users authenticate once with a trusted identity provider and can securely access protected resources without maintaining separate login credentials for the application. The exercise highlights how centralized authentication improves security, simplifies user management, and provides a seamless login experience in modern microservice-based applications.

Outputs:

1. Postman output1



2. Postman output2

The image shows the Postman interface for a GET request to `http://localhost:8080/employees`. The response status is **200 OK** with a response time of 165 ms and a body size of 18.59 KB. The response body is in JSON format, showing an array of employee objects. The first object is for an employee with id 1, name John, salary 75000.0, permanent true, and a date of birth of 1998-05-09T18:30:00.000Z. The second object is for an employee with id 3.

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body

200 OK • 165 ms • 18.59 KB • [Icons]

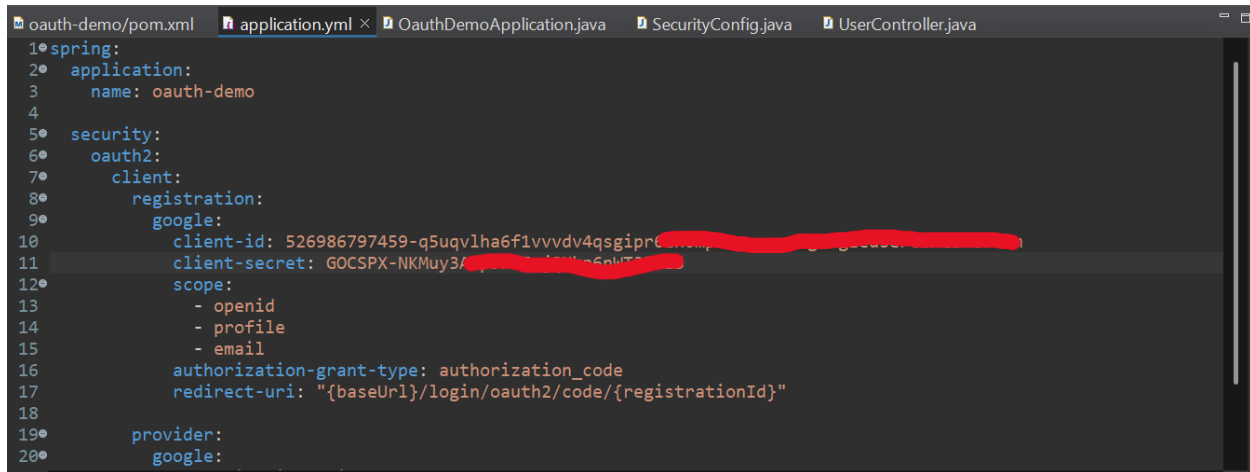
{ } JSON [Preview] [Visualize]

```
1 [
2   {
3     "id": 1,
4     "name": "John",
5     "salary": 75000.0,
6     "permanent": true,
7     "dateOfBirth": "1998-05-09T18:30:00.000Z",
8     "skillList": [
9       {
10        "id": 3,
```

3. Console

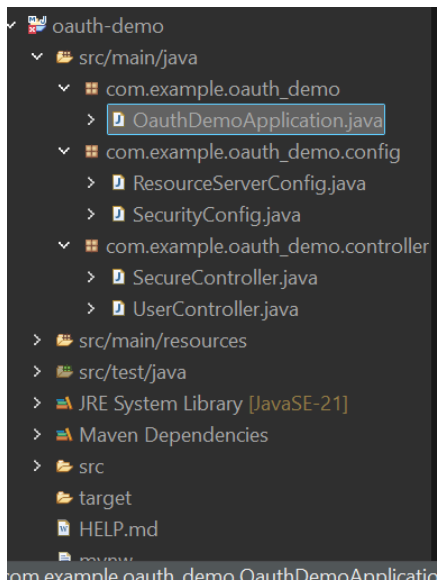
```
lication      : Starting OauthDemoApplication using Java 22.0.2 with PID 19336 (C:\Users\HP PRO BOOK 440 G6\ec1
lication      : No active profile set, falling back to 1 default profile: "default"
rver          : Tomcat initialized with port 8080 (http)
dardService   : Starting service [Tomcat]
dardEngine    : Starting Servlet engine: [Apache Tomcat/11.0.22]
xtInitializer  : Root WebApplicationContext: initialization completed in 2486 ms
rver          : Tomcat started on port 8080 (http) with context path '/'
lication      : Started OauthDemoApplication in 6.65 seconds (process running for 12.177)
```

4. Application.yml:



```
1 spring:
2   application:
3     name: oauth-demo
4
5   security:
6     oauth2:
7       client:
8         registration:
9           google:
10            client-id: 526986797459-q5uqv1ha6f1vvvdv4qsgipr6...
11            client-secret: GOCSPX-NKMuy3A...
12            scope:
13              - openid
14              - profile
15              - email
16            authorization-grant-type: authorization_code
17            redirect-uri: "{baseUrl}/login/oauth2/code/{registrationId}"
18
19       provider:
20         google:
```

5. Project folder:



```
✓ oauth-demo
├── src/main/java
│   ├── com.example.oauth_demo
│   │   ├── OAuthDemoApplication.java
│   │   ├── com.example.oauth_demo.config
│   │   │   ├── ResourceServerConfig.java
│   │   │   ├── SecurityConfig.java
│   │   │   └── com.example.oauth_demo.controller
│   │   │       ├── SecureController.java
│   │   │       └── UserController.java
│   ├── src/main/resources
│   ├── src/test/java
│   ├── JRE System Library [JavaSE-21]
│   ├── Maven Dependencies
│   ├── src
│   ├── target
│   ├── HELP.md
│   └── mvnw
└── com.example.oauth_demo.OAuthDemoApplication
```